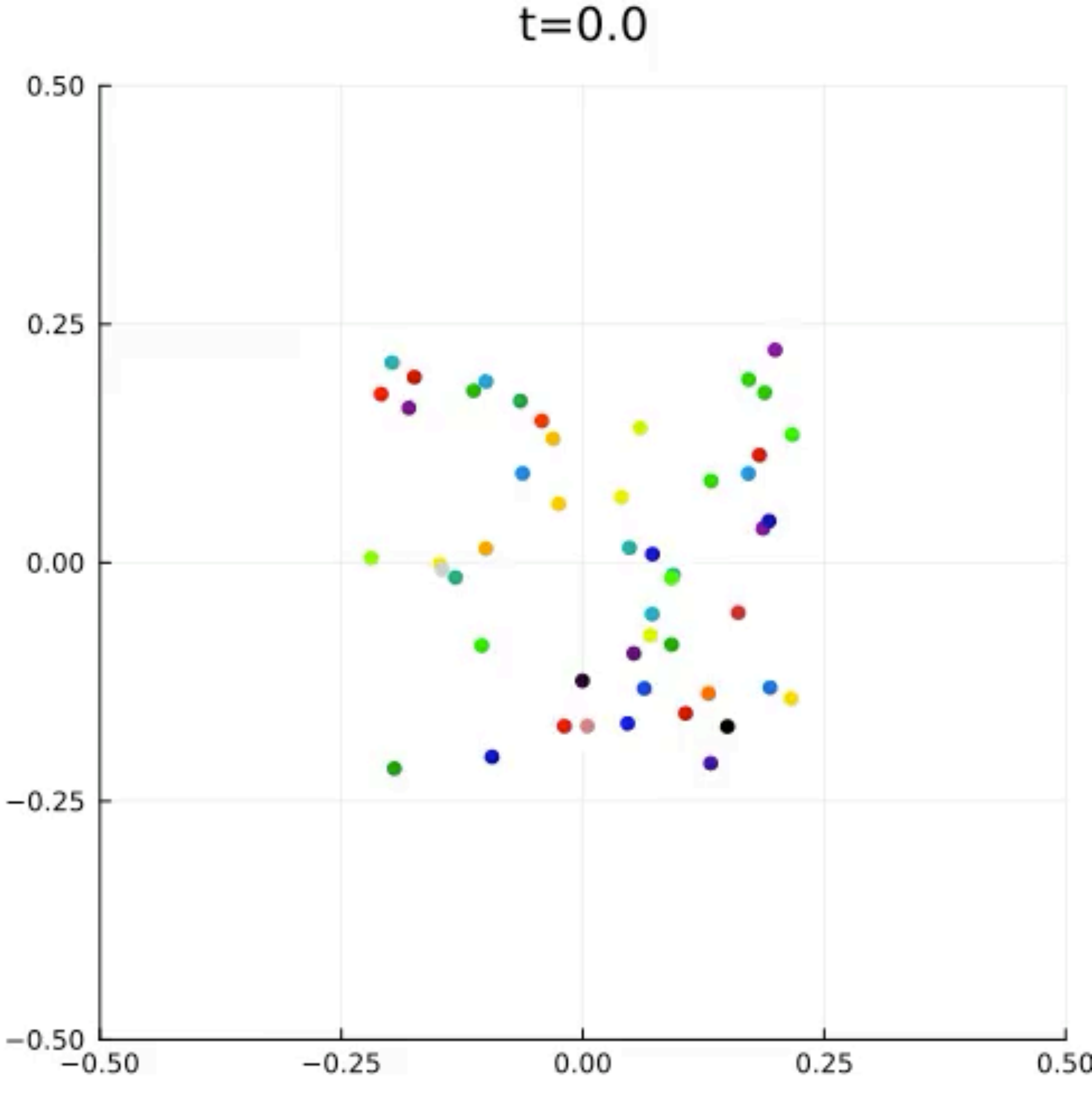
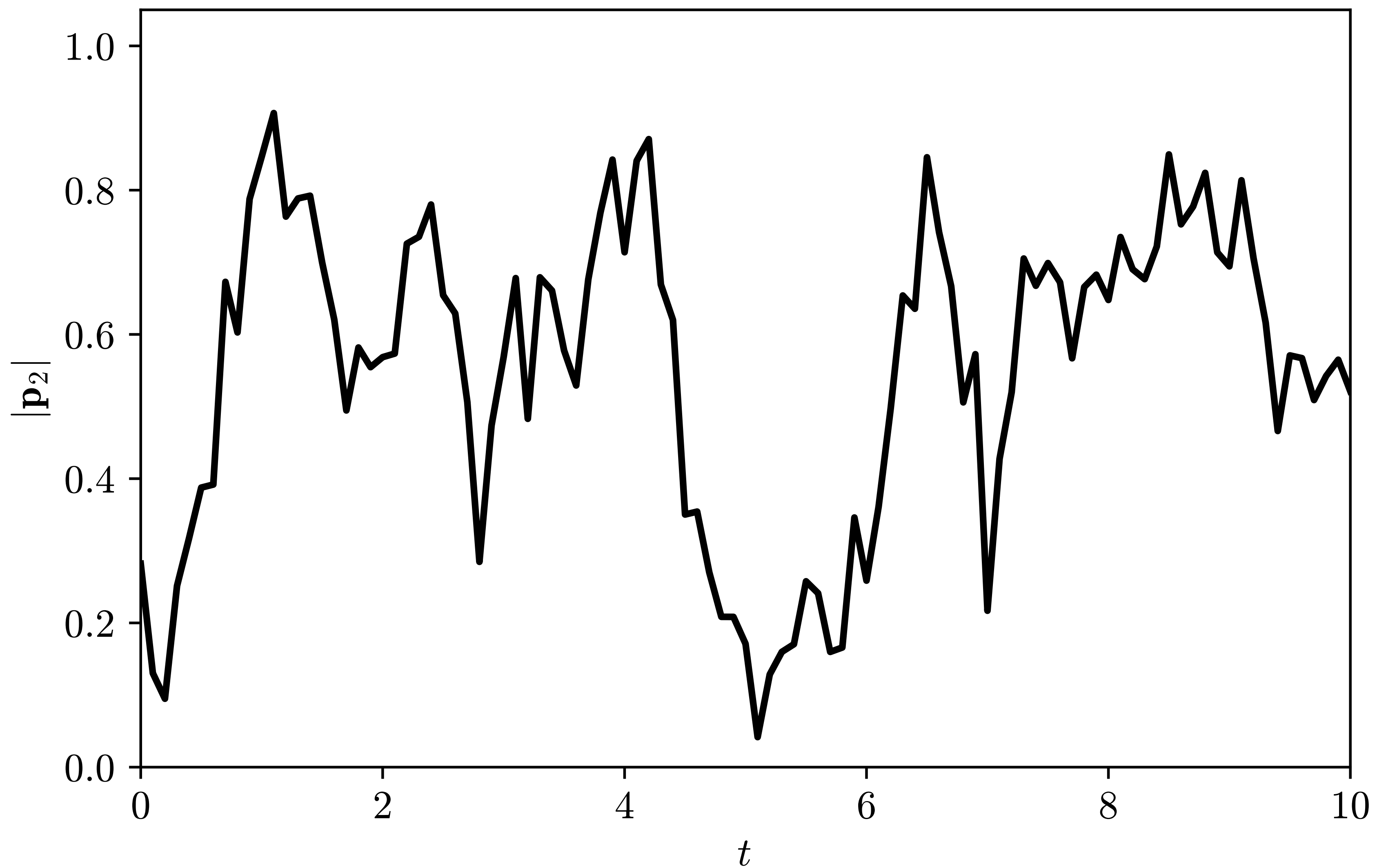
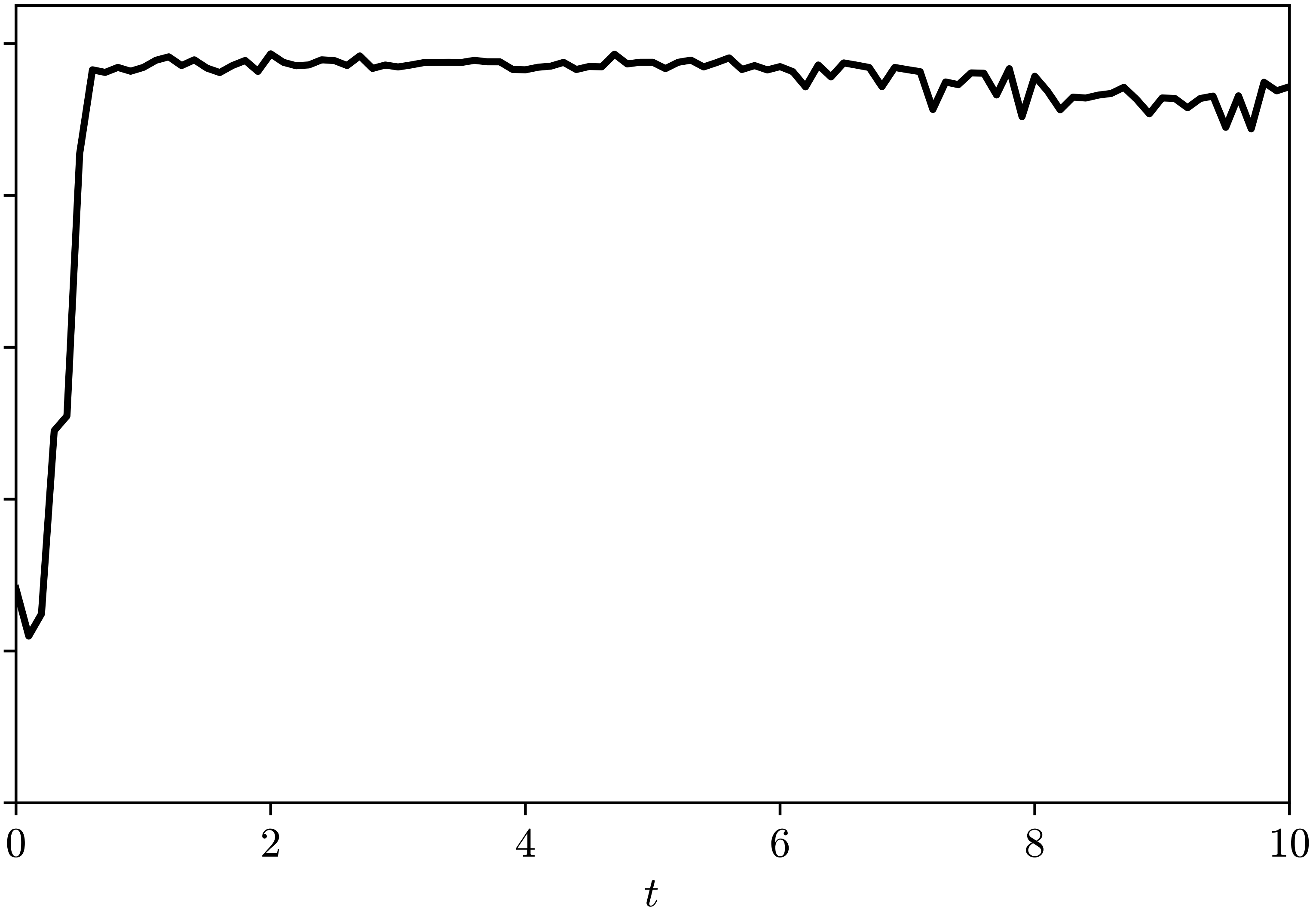
- Fluctuations and dynamical metastability for $\tau \approx \tau^*$ for finite N?
- $|\mathbf{p}_2| = \left| \frac{1}{N} \sum_{i=1}^{N} v(2\theta_t^i) \right|$


t=0.0
0.50
0.25
0.00
−0.25
−0.50
−0.50
−0.25
0.00
0.25
0.50



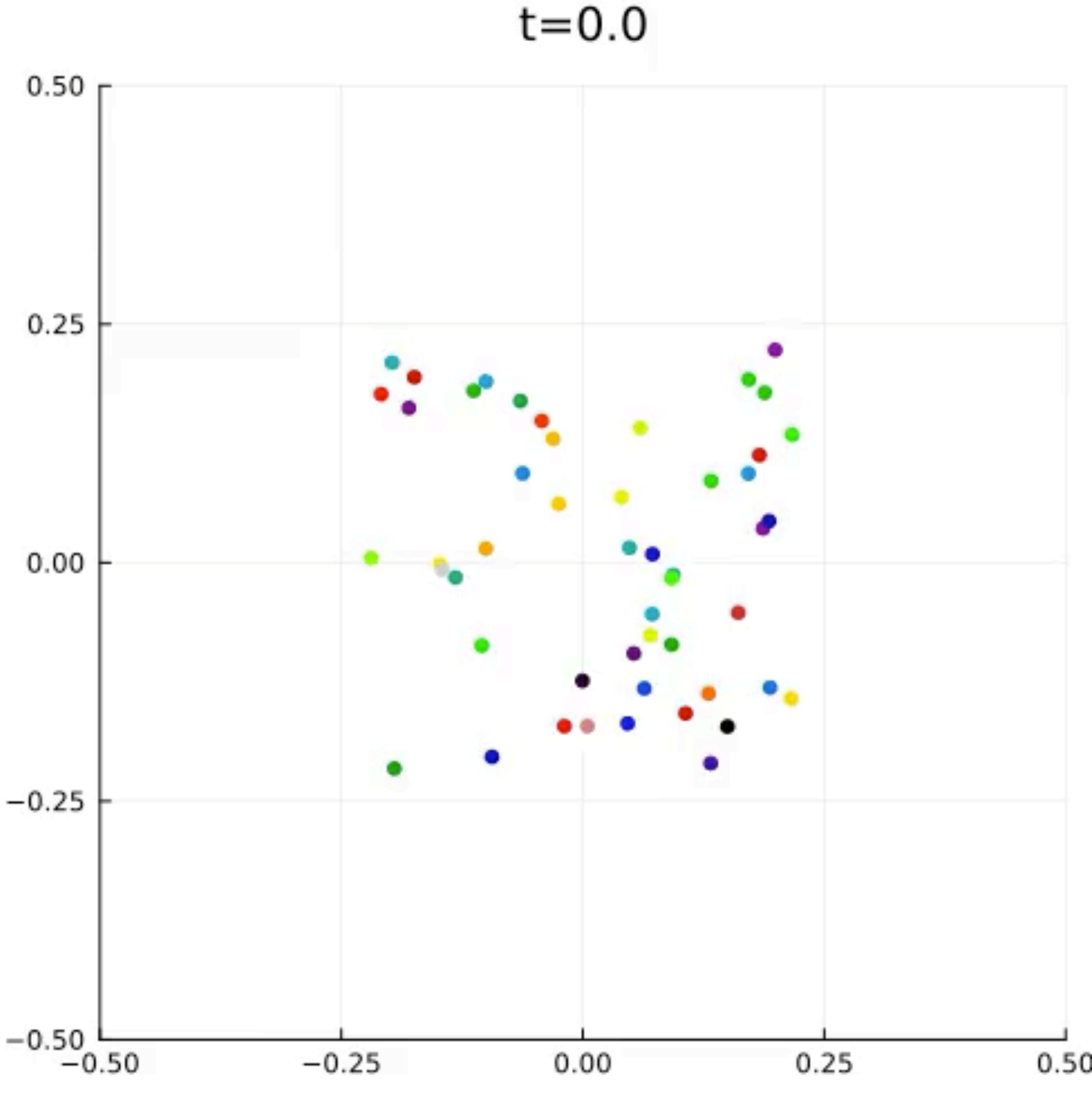
1.0
0.8
0.6
0.4
0.2
0.0
$|\mathbf{p}_2|$
0
2
4
6
8
10
t

$$\tau \approx \tau^*$$


0
2
4
6
8
10
t

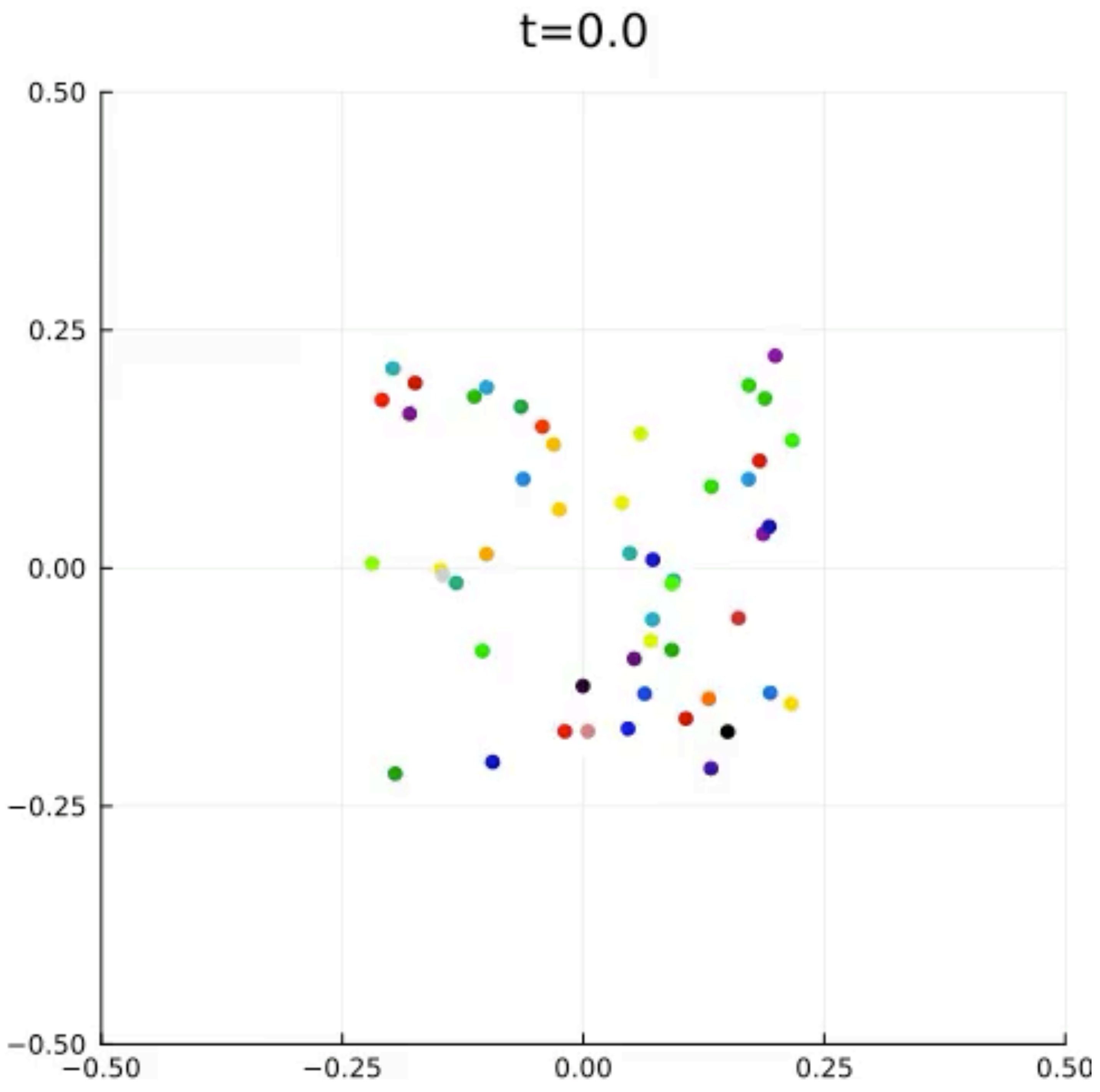
$$\tau > \tau^*$$

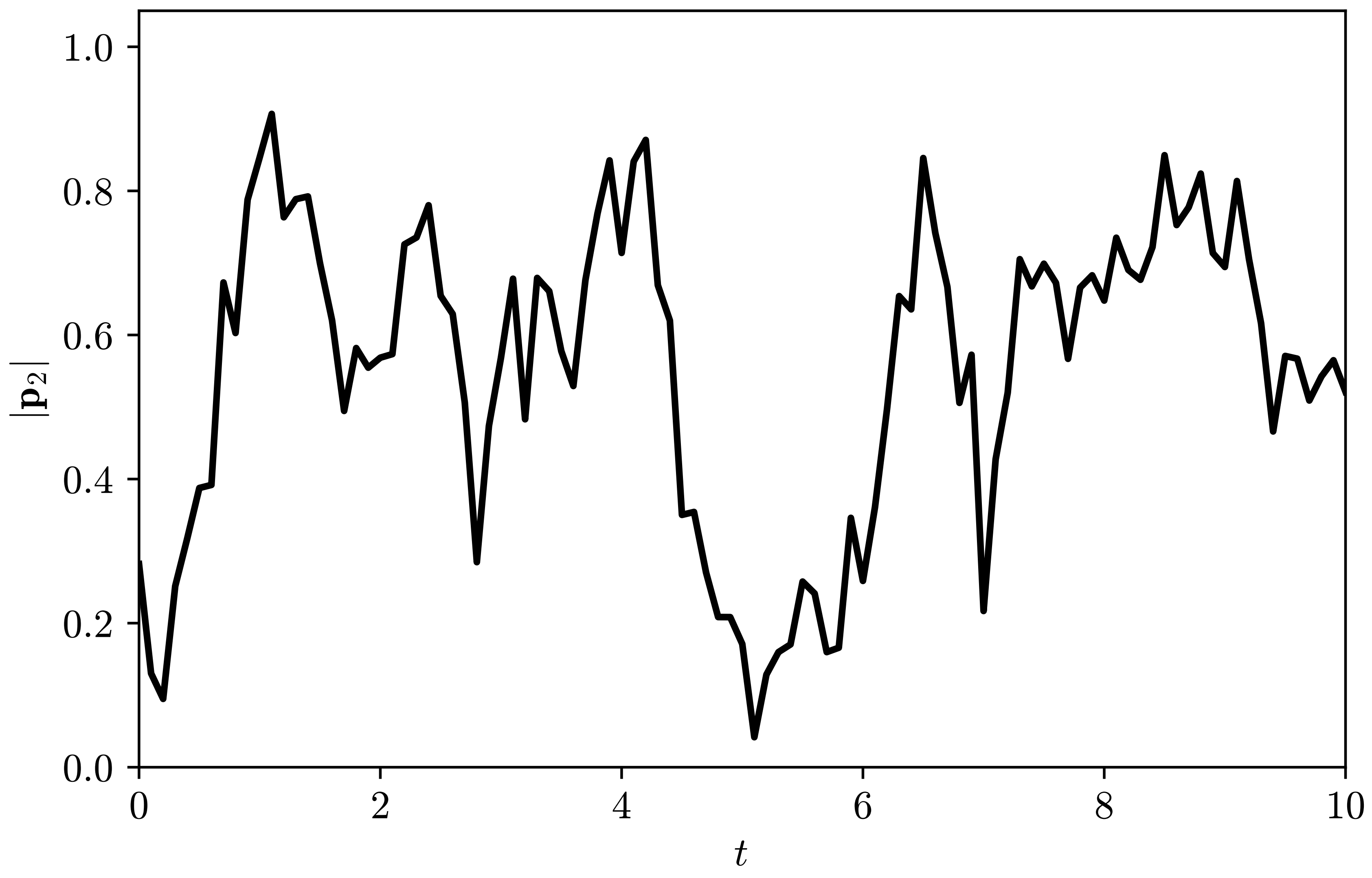
Active ants: metastability? (Open)


t=0.0
0.50
0.25
0.00
−0.25
−0.50
−0.50
−0.25
0.00
0.25
0.50

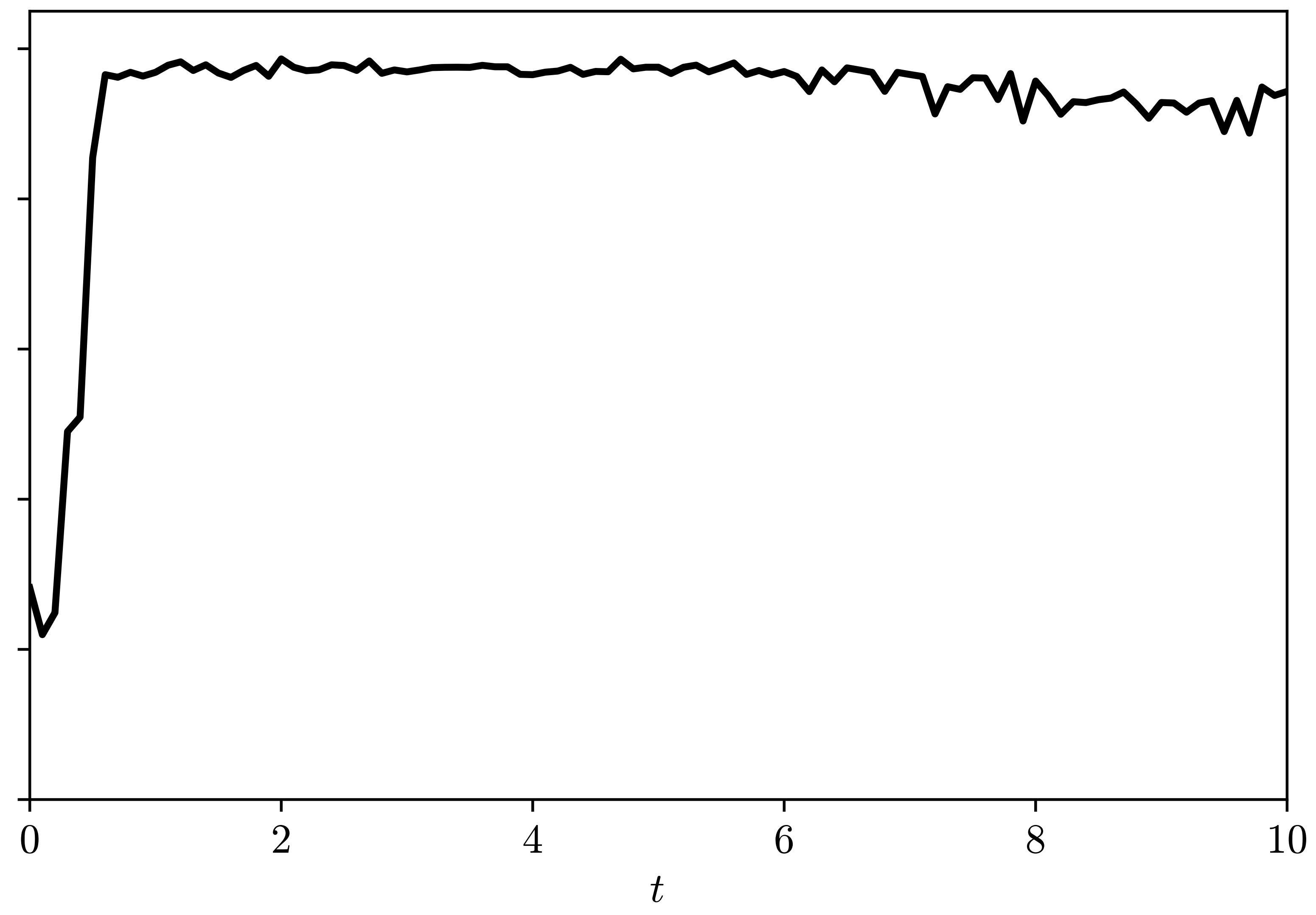
Active ants: metastability? (Open)

- Fluctuations and dynamical metastability for $\tau \approx \tau^*$ for finite N?
- $|\mathbf{p}_2| = \left| \frac{1}{N} \sum_{i=1}^{N} v(2\theta_t^i) \right|$





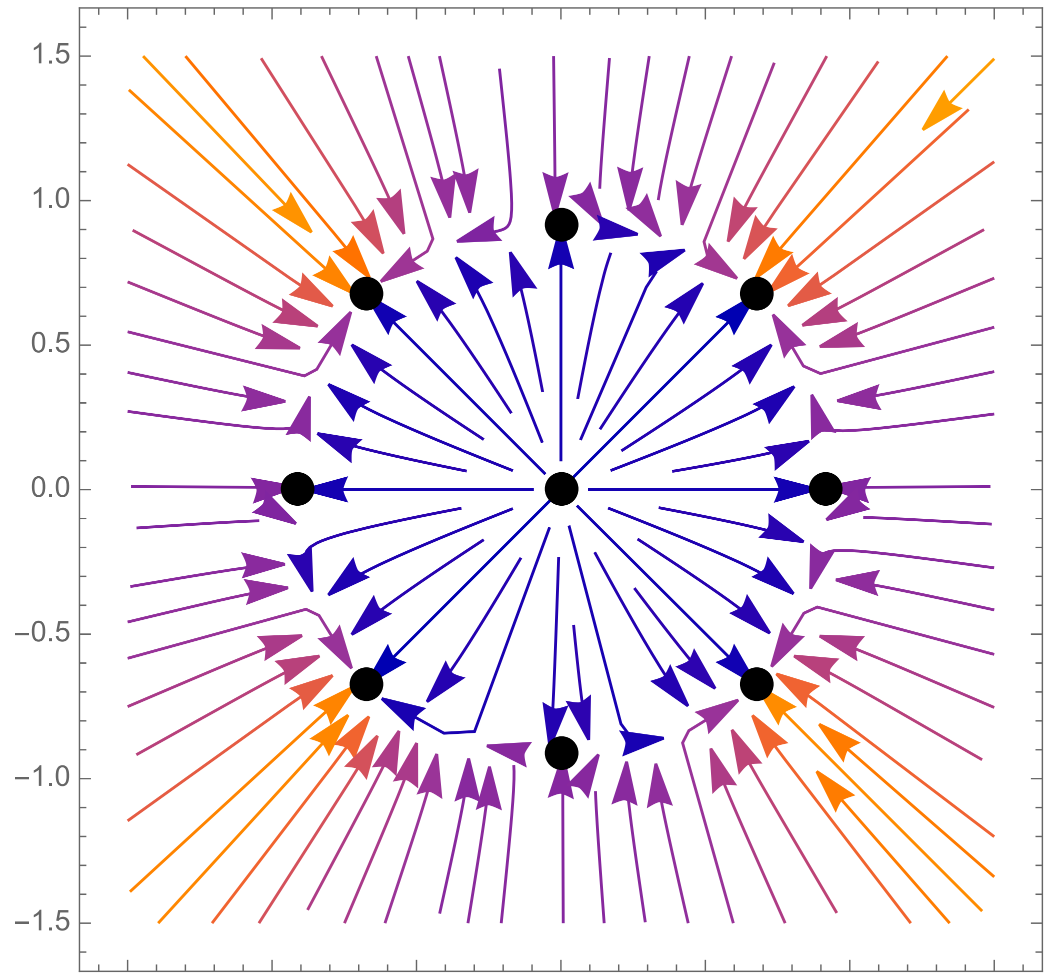
$\tau \approx \tau^*$



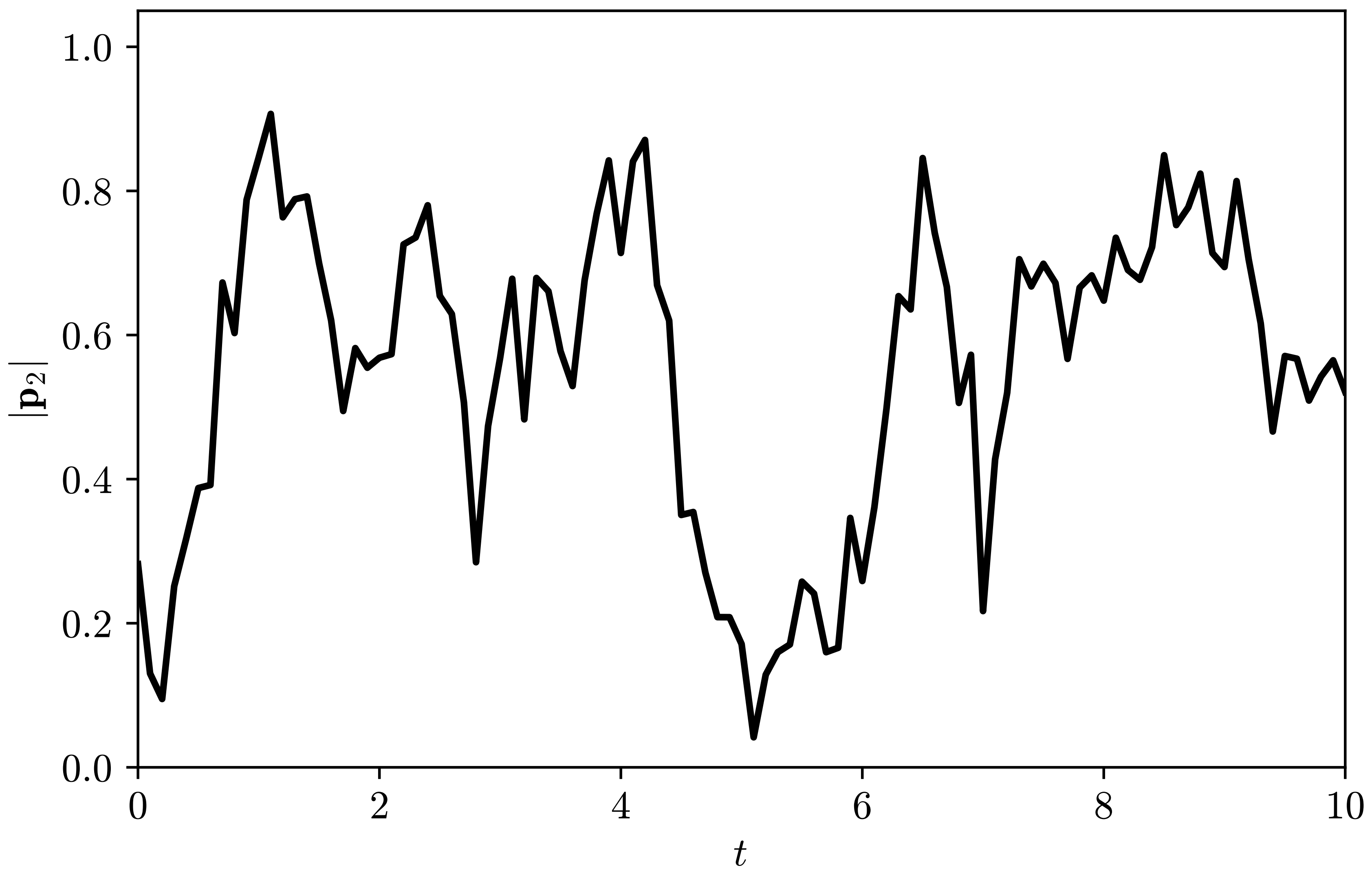
$\tau > \tau^*$

Active ants: metastability? (Open)

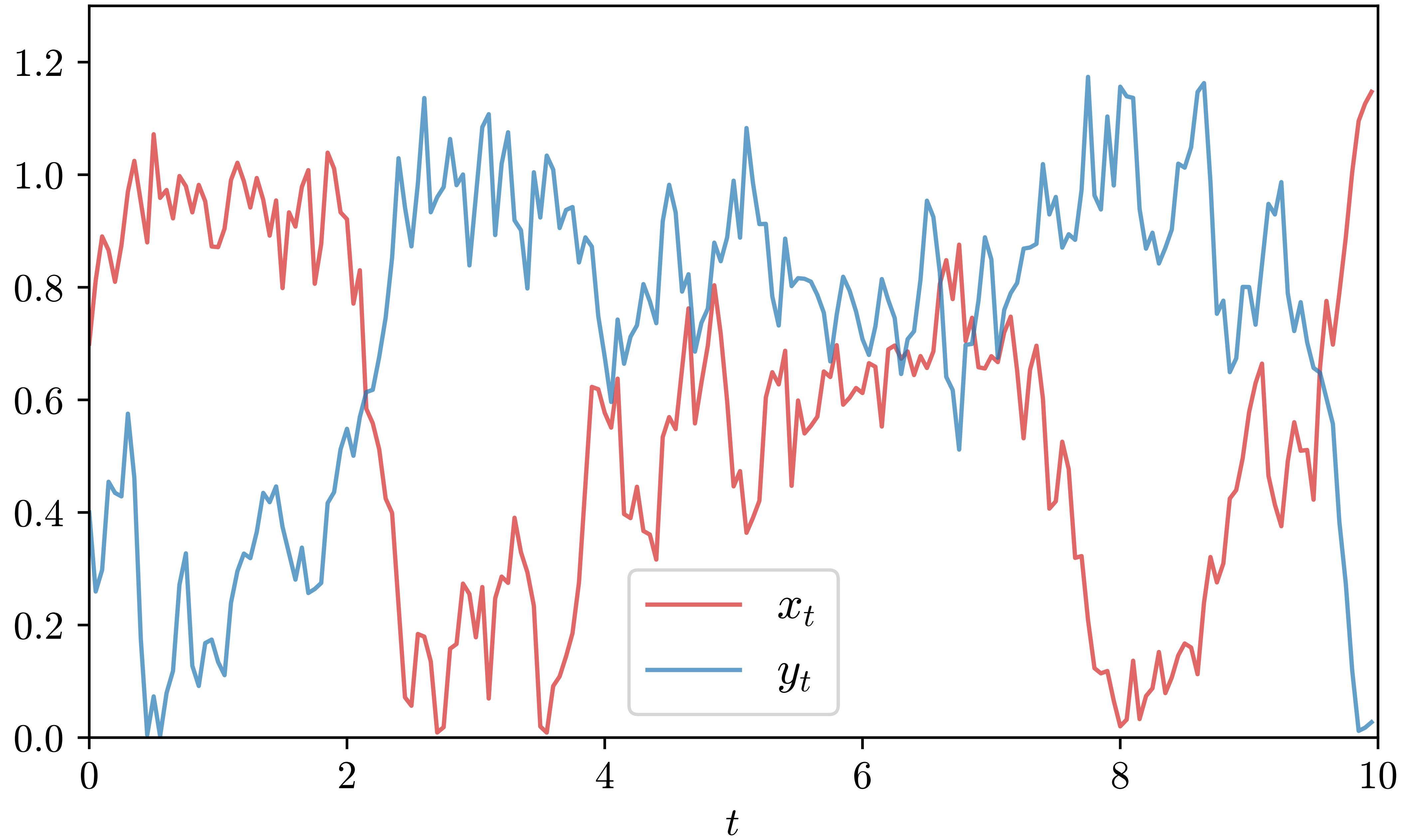
- $|\mathbf{p}_2| = \left| \frac{1}{N} \sum_{i=1}^{N} v(2\theta_t^i) \right|$
- Compare with a Brownian particle (x_t, y_t) following the ODE vector field



$\tau < \tau^*$



$\tau \approx \tau^*$



$\tau = \tau^*$